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Effects of music on the treatment of attention deficit and hyperactivity disorder (ADHD) in children and adolescents: a systematic review of current evidence

Efeitos da música no tratamento do transtorno do déficit de atenção e hiperatividade (TDAH) em crianças e adolescentes: uma revisão sistemática das evidências atuais

Efectos de la música en el tratamiento del trastorno por déficit de atención e hiperactividad (TDAH) en niños y adolescentes: una revisión sistemática de la evidencia actual

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Music therapy improves concentration, self-esteem, social skills, and hyperactivity control in ADHD patients, offering a viable complementary treatment to traditional pharmacological approaches.

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This systematic review synthesizes recent evidence (2014–2024) on music's therapeutic role in ADHD, highlighting its methodological rigor and practical relevance for non-invasive interventions.

ABSTRACT

Background: attention deficit hyperactivity disorder (ADHD) is a mental condition that encompasses a compendium of persistent difficulties, such as difficulty concentrating, hyperactivity, and impulsive behavior, which is why musical activities are currently used to address it. **Methods:** this study is based on a systematic review to analyze the effects of music in the management of attention deficit hyperactivity disorder (ADHD) in children and adolescents. The exploration was supported by Boolean operators in ERIC, Scopus, PubMed, and Google Scholar databases, considering academic articles and technical documents endorsed by specialists, published between 2014 and 2024 in English and Spanish. **Results:** starting from 756 articles initially collected. In the initial review, selection criteria were applied, which allowed 24 documents to be specified for the final analysis; **Conclusions:** among the main effects of music on ADHD, it can be treated as a non-pharmacological therapy that improves the level of concentration, raises self-esteem, intervenes in the progressive development of social skills, and promotes control of hyperactivity and directs people towards reducing anxiety states.

Keywords: music; ADHD; self-esteem; social skills; anxiety.

RESUMO

Antecedentes: o transtorno de déficit de atenção e hiperatividade (TDAH) é uma condição mental que engloba um compêndio de dificuldades persistentes, como dificuldade de concentração, hiperatividade e comportamento impulsivo, razão pela qual atualmente atividades musicais são utilizadas para enfrentá-lo. **Métodos:** este estudo baseia-se em uma revisão sistemática com o objetivo de analisar os efeitos da música no manejo do transtorno de déficit de atenção e hiperatividade (TDAH) em crianças e adolescentes. A exploração contou com o apoio de operadores booleanos, nas bases de dados ERIC, Scopus, PubMed e Google Scholar, considerando artigos acadêmicos e documentos técnicos endossados por especialistas, publicados entre 2014 e 2024 em inglês e espanhol. **Resultados:** a partir de 756 artigos coletados inicialmente. Na revisão inicial foram aplicados critérios de seleção, o que permitiu especificar um total de 24 documentos para a análise final. **Conclusões:** Concluindo, entre os principais efeitos da música no TDAH, ela pode ser tratada como uma terapia não farmacológica que melhora o nível de concentração, eleva a autoestima, intervém no desenvolvimento progressivo das habilidades sociais e promove o controle da hiperatividade e direciona as pessoas para a redução dos estados de ansiedade.

Palavras-chave: música; TDAH; autoestima; habilidades sociais; ansiedade.

RESUMEN

Antecedentes: el trastorno por déficit de atención e hiperactividad (TDAH) es una afección mental que abarca un compendio de dificultades persistentes, como la dificultad de concentración, hiperactividad y conducta impulsiva, por lo que actualmente se emplean actividades musicales para su abordaje. **Métodos:** este estudio se fundamenta en una revisión sistemática, con el objetivo de analizar los efectos de la música en el manejo del trastorno por déficit de atención e hiperactividad (TDAH) en niños y adolescentes. La exploración se sustentó en el apoyo de operadores booleanos, en las bases de datos ERIC, Scopus, PubMed y Google Scholar, considerando artículos académicos y documentos técnicos avalados por especialistas, publicados entre 2014 y 2024, tanto en inglés como en español. **Resultados:** Inicialmente, se recopilaron 756 artículos. En la revisión inicial, se aplicaron criterios de selección, lo que permitió precisar un total de 24 documentos para el análisis final. **Conclusiones:** Se llegó a la conclusión de que la música puede ser usada como una terapia no farmacológica en el TDAH, ya que propicia el mejoramiento del nivel de concentración, eleva la autoestima, interviene en el desarrollo progresivo de habilidades sociales, favorece el control de la hiperactividad y ayuda a que las personas reduzcan la ansiedad.

Palabras clave: música; TDAH; autoestima; habilidades sociales; ansiedad.

INTRODUCTION

Nowadays, music is the object of study due to the way it facilitates the study of brain functioning (Reybrouck et al., 2021). In this context, music is used as a vehicle that drives response actions in acoustically conditioned environments, facilitating the observation of human emotions (Blasco-Magraner et al., 2021). However, a specific area in the brain that allows music processing, and the configuration of neural networks has not yet been identified (Fasano, 2022).

However, it has been specified that there is a series of connections between different regions within the brain, which allow us to understand music as it is perceived (Chen et al., 2021). In this sense, listening to a musical melody generates a wide variety of activities and brain stimuli that directly influence important aspects of life such as mood, stress control, anxiety, or fatigue (Loui, 2020).

Therefore, the human being has seen himself as an extremely complex system, that coexists under synchronized interactions with various subsystems of biophysical, biochemical, biological, neurological, and emotional order, which provide everyone with a configuration of reactions or responses to stimulating events of internal and external origin (Simor et al., 2020). In this sense, from the very moment of conception, the human embryo initiates a progressive transformation process, where the genetic contributions of its parents react and transform it into a being with very special characteristics (Rusca-Jordán & Cortez-Vergara, 2020).

In this order of ideas, one of the aspects that implies a process of meticulous attention from health professionals is represented by the development of the complex neurological system of the individual (Eichmüller & Knoblich, 2022). This aspect maintains the progressive attention of the specialists in the area, with the observation of the integral development of the future neonate, on possible alterations or limitations in its neurological development (Bakhsh et al., 2023). In this particular, one of the aspects of greatest attention in the last five years has been the detection of attention deficit hyperactivity disorder (ADHD) (Haque et al., 2023).

ADHD is a neurobiological disorder that manifests preponderantly in childhood, but in some cases can persist into adolescence and adulthood (Da Silva et al., 2023). This disorder mainly affects a person's ability to pay attention, control impulses, and have regular mastery over physical activity (Jones et al., 2023).

From the scientific perspective, the probability that ADHD has its beginnings during the fetal development of an individual has aroused great interest in both genetic and environmental factors associated with the parents (Mahrous et al., 2024), to establish methods of prevention and care with emerging protocols that precede the presence and installation of the syndrome (Miller et al., 2023). Although it is true that so far it has not been possible to establish a standardized curative action, the existence of certain disruptive strategies that have generated positive responses is known, evidencing changes in the social behavior of individuals with ADHD (Langberg et al., 2023; Rodwin et al., 2023).

In this order of ideas, Rusca-Jordan & Cortez-Vergara (2020) confirm that this condition is detected through a clinical diagnosis always considering that, by its nature, the symptoms can be observed developmentally. However, its confirmation can be supported using international diagnostic manuals, such as the ICD-10 of the World Health Organization or the DSM-5 of the American Psychiatric Association. Similarly, every child or adolescent should have a comprehensive individualized treatment plan that highlights the chronicity and impact of the condition, involving psychopharmacological actions, behavioral therapies, or emergent activities such as the use of music therapy.

In the same context, Leon et al. (2024) mentions the use of interventions in specific groups of children and adolescents with music therapy and nerve stimulation as a complement to established pharmacological actions, which are associated with significant improvements in attention, memory, and executive function. Similarly, Cibrian et al. (2022) mention music therapy augmented with the use of electronic and digital devices to counteract the effects of ADHD, which is a strategy aimed at the socio-communicative and emotional regulation of children and adolescents.

Based on the above, the present study is based on a systematic review whose objective is to analyze the effects of music in the management of attention deficit hyperactivity disorder (ADHD) in children and adolescents. For this purpose, the following research question has been established: in what way does the use of music favor the management of attention deficit hyperactivity disorder (ADHD) in children and adolescents? The objective is to obtain updated information on strategies based on music therapy that facilitate the improvement of the performance of young university students.

Objective

This study is based on a systematic review that complies with methodological aspects and Prisma (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. The general objective was to analyze the effects of music in the management of ADHD in children and adolescents.

METHODS

A thorough literature search was carried out in various electronic databases that have published studies with profiles related to mental health, social sciences, and educational sciences, such as ERIC, Scopus, PubMed, and Google Scholar. The search strategies applied are shown in detail in Table 1, where Boolean operators, descriptors, or keywords related to "ADHD", "Concentration level", "Self-esteem", "Social skills", "Hyperactivity" and "Anxiety" are included.

Table 1. Database search equations

Database	Database search
ERIC	"Music" AND "Music therapy" OR "Music therapy" AND "Attention deficit" OR "Hyperactivity" OR "ADHD" AND "Concentration level" AND "Self-esteem" AND "Social skills" AND "Hyperactivity" AND "Anxiety"
Scopus	TITLE-ABS-KEY "Music" AND "Music therapy" OR "Music therapy" AND "Attention deficit" OR "Hyperactivity" OR "ADHD" AND "Concentration level" AND "Self-esteem" AND "Social skills" AND "Hyperactivity" AND "Anxiety"
PubMed	"Música" AND "Musicoterapia" OR "Terapia musical" AND "Déficit de atención" OR "Hiperactividad" OR "TDAH" AND "Nivel de concentración" AND "Autoestima" AND "Habilidades sociales" AND "Hiperactividad" AND "Ansiedad"
Google Scholar	"Música" AND "Musicoterapia" OR "Terapia musical" AND "Déficit de atención" OR "Hiperactividad" OR "TDAH" AND "Nivel de concentración" AND "Autoestima" AND "Habilidades sociales" AND "Hiperactividad" AND "Ansiedad"
	"Music" AND "Music therapy" OR "Music therapy" AND "Attention deficit" OR "Hyperactivity" OR "ADHD" AND "Concentration level" AND "Self-esteem" AND "Social skills" AND "Hyperactivity" AND "Anxiety"

Note. Authors' development

Each study identified was examined in detail according to the pre-established inclusion and exclusion criteria, which can be seen in Table 2. These were designed to guarantee the consideration of original, new, and novel articles, included in indexed scientific journals that had been peer-reviewed and that evidenced the approach to the subject matter of the study. The review was directed to papers published in both English and Spanish, from 2014 to 2024. Commentaries, letters to the editor, degree theses, master's theses, and any other type of publication that was not an original research article were excluded.

Table 2. Inclusion and exclusion criteria

Inclusion criteria	Exclusion criteria
Original academic articles	Master's and doctoral dissertations, theses, and letters from editors
Published in peer-reviewed journals	Non-indexed papers
Addressing the effects of music in the management of ADHD in children and adolescents	Unrelated topics on the effects of music in the management of ADHD in children and adolescents. Related work but applied to college student populations
English or Spanish language studies	Languages other than English or Spanish.
From 2014 to 2024	Before 2014

Note. Authors' development

After eliminating duplicate papers, the eligibility of the titles and abstracts of the identified studies was independently assessed. In case of doubt or insufficient information, the full text of the article was obtained for further assessment. The methodological quality of the included studies was assessed using the risk of bias assessment tools suitable for different study designs.

Highlights of each study were specified as details on the overall objective, study methodology, main results, and the contribution generated. The data were systematically compiled and structured in an evidence table to facilitate synthesis and analysis of the results, which were systematized and synthesized in a narrative fashion, exposing convergent and divergent points between studies, and possible sources of heterogeneity were explored. Finally, a critical analysis of the strengths and limitations of the existing evidence was applied.

RESULTS AND DISCUSSION

This The search in the databases yielded a total of 756 records. After removing 249 duplicates, it was reduced to 507 studies, of which 399 were excluded after reviewing titles and abstracts for not meeting the eligibility criteria. The remaining 108 articles were assessed in full text, and 84 were excluded due to irrelevance. Finally, 24 articles met all the inclusion criteria and were included in the systematic review. Table 3 describes the information flow process generated.

Table 3. Stages developed in the collection of information, according to the Prisma Flow

Stages	No. of records
Records of identified starts in databases	756
Records after discarding duplicates	507
Records excluded by title/abstract	399
Full-text papers screened for eligibility	108
Full-text articles not selected	84
Articles included in qualitative synthesis	24

Note. Authors' development

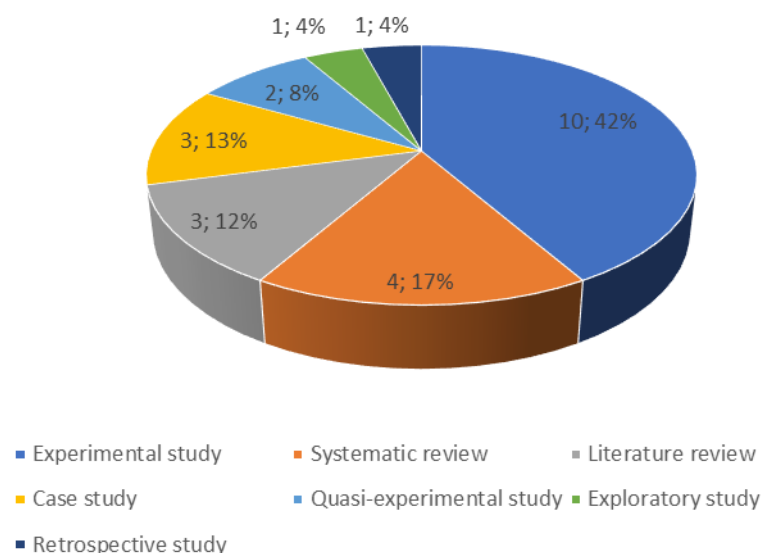
According to the relevant data obtained from the review, 54% of the articles are registered in the Scopus database, 21% in Google Scholar, 17% in PubMed, and the remaining 8% in the ERIC academic education database. In addition, it was possible to verify that in the study period, the year with the highest frequency of publications was 2023, with a total of 6 articles reviewed. Table 4 shows the above information in detail.

Table 4. Publications analyzed by database and year of publication

Database	Publication date											Total	%
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030		
ERIC	0	0	0	0	1	0	0	0	1	0	0	2	8 %
PubMed	0	0	0	0	0	0	0	0	0	0	3	4	17 %
Google Scholar	1	0	2	0	0	0	0	0	0	2	0	5	21 %
Scopus	1	0	1	0	0	1	2	2	3	1	2	13	54 %
Total	2	0	3	0	1	1	2	3	3	6	3	24	100 %

Note. Authors' development

Concerning the type of study, it was possible to specify, according to the methodology described in each case, that 42% is framed within the experimental design, 17% have the characteristics of a systematic review, 13% are literature reviews and case studies, 8% are quasi-experimental design studies and, finally, in equal parts, 4% are exploratory and retrospective studies. Figure 1 shows the percentage distribution of the information.

Figure 1. Percentage distribution according to the type of study detected in the reviewed academic articles

Note. Authors' development

Table 5 shows the relevant findings of the 24 articles included in the systematic review. These address various constructs that could be directly and indirectly associated with the effects of music in the treatment of attention deficit hyperactivity disorder (ADHD) in college students. This informative ordering facilitates an overview of the available evidence, as well as the comparison and synthesis of the results of the different studies.

Table 5. Qualitative screening results

Author(date)	Title	Finding
Situmorang <i>et al.</i> (2018)	Comparison of the Effectiveness of CBT Group Counseling with Passive vs Active Music Therapy to Reduce Millennials Academic Anxiety	The results showed that during pretest versus posttest cognitive behavioral therapy (CBT) with passive therapy was more effective in reducing academic anxiety compared to active music therapy. At pretest versus posttest, active was more effective in reducing academic anxiety compared to passive music therapy.
Jacob & Pillay (2021).	Effectiveness of music therapy on reading skills of pupils with intellectual disability	Music therapy improves the reading skills of students with intellectual disabilities.
Mendes <i>et al.</i> (2024)	Effects of Background Music on Attentional Networks of Children With and Without Attention Deficit/Hyperactivity Disorder: Case Control Experimental Study	In the experiment applied to a total of 76 children (34 with ADHD and 42 with normal development) no direct effect of music was found, and interference in the attentional network of children and adolescents could not be verified. Future studies will be necessary to validate this.
Martin-Moratinos <i>et al.</i> (2023)	Effects of Music on Attention-Deficit/Hyperactivity Disorder (ADHD) and Potential Application in Serious Video Games: Systematic Review	Of the 70 records identified, 17 provided findings that music may be beneficial in various domains of ADHD.
Park <i>et al.</i> (2023).	Effects of music therapy as an alternative treatment for depression in children and adolescents with ADHD by activating serotonin and improving stress coping ability	The application of music therapy as an alternative treatment for children and adolescents with ADHD showed positive neurophysiological and psychological effects.
Lee <i>et al.</i> (2024)	Music and movement therapy improves quality of life and attention and associated electroencephalogram changes in patients with attention-deficit/hyperactivity disorder.	The results revealed that the participant's quality of life increased significantly after the 8-week intervention. EEG analysis demonstrated an increase in alpha power and Higuchi fractal dimension, and a decrease in delta power in certain EEG channels.
Llamas (2014).	Alumnos con TDAH y musicoterapia: cómo trabajar en primaria para mejorar el desarrollo personal y escolar en estos niños.	The intervention through music therapy favored the progressive incorporation of children with ADHD into the work team, getting along better with their peers and improving their school performance.
Khaleghian <i>et al.</i> (2023).	The Effectiveness of Music Therapy on Social Skills and Psychological Health of Students with Attention Deficit Hyperactivity Disorder	ANCOVA results showed that music therapy increases the social skills and psychological health of hyperactive students, such as those diagnosed with ADHD.
Hemmati <i>et al.</i> (2016)	The effect of group art therapy on the self-restraint of students with attention-deficit/hyperactivity disorder	Data analyses using ANCOVA and MANCOVA showed that there was a significant increase in the mean scores of self-controls and subscales (aggression suppression, impulse control, consideration of others, and responsibility) in the experimental group.
Kim & Wheeler (2023)	Analyzing the Role of Melodic Contours in Music Therapy for ADHD Therapeutic Interventions	Music therapy can be used to address attention, impulsivity, and emotional regulation in patients. This review warrants further research in clinical trials to test the efficacy of using specific melodic contours for the treatment of ADHD.
Acebes-De Pablo & Carabias-Galindo, (2016)	El alumnado de primaria con trastorno por déficit de atención/hiperactividad (TDAH): la musicoterapia como herramienta integradora dentro del contexto del aula de música	After the intervention, there is evidence of some improvement in the symptoms of the disorder and the learning level of the study subjects.
Madjar <i>et al.</i> (2020)	Contrasting effects of music on reading comprehension in preadolescents with and without ADHD	The findings indicated an elevation of reading comprehension and a variation in participants' heart rate due to the effects of musical conditions. This suggests that music may improve the attention skills of preadolescents with ADHD. Still, it suggests the need to identify an optimal fit between individual and contextual characteristics.
Shuman <i>et al.</i> (2016).	Group music therapy impacts the mood states of adolescents in a psychiatric hospital setting.	Based on a binary outcome model of change or no change, the probability of a change in total mood score was high (probability of change 0.82 [95 % CI 0.79, 0.85]). The probability of having a change in mood is high and statistically and clinically significant. It appeared that the intervention worked equally well for patients of different genders, ages, diagnoses, and acuity levels.
Martínez-Vérez <i>et al.</i> (2024)	Interventions through Art Therapy and Music Therapy in Autism Spectrum Disorder, ADHD, Language Disorders, and Learning Disabilities in Pediatric-Aged Children: A Systematic Review	Among the main findings are that art therapy and music therapy have a significant impact on symptomatology, behavior, and communication, as well as social, cognitive, and emotional skills in the pediatric populations studied with ADHD.
Acebes-De Pablo & Giraldez-Hayes (2019)	El papel de la musicoterapia y las terapias alternativas en el tratamiento del TDAH: un estudio exploratorio	The results suggest that the application of alternative therapies is recommended as a complementary treatment and that music therapy is equally recommended but less known by professionals working with ADHD.
Chenguang (2021)	Effects of Music Therapy combined with Cognitive Behavioral Intervention on the Cognitive Ability of children with Attention Deficit Hyperactivity Disorder	After the applied intervention, the test results had a significant change in the experimental group. In the observation group, the percentage of conceptual level and the number of completed lessons had significantly increased, and the number of discontinuous errors and the number of continuous errors after the intervention had drastically decreased compared to the previous ones.
Dursun <i>et al.</i> (2021).	Probable role of listening therapy in the management of ADHD symptoms: Three case studies	Instruments such as the CPT (MOXO d-CPT), the Conners Rating Scales for Parents (CPRS) and Teachers (CTRS), and the Adult ADHD Self-Report Scale (ASRS) were used according to the ages of the participants, in addition to the listening checklist. Although the differences in scores did not represent any statistical significance, the findings may still be noteworthy indicators for future prospective studies.
Luo <i>et al.</i> (2023).	Yoga and music intervention reduces inattention, hyperactivity/impulsivity, and oppositional defiant disorder in children consumers with comorbid ADHD and ODD.	The combined yoga and music intervention had a positive effect on inattention and hyperactivity/impulsivity in children with ADHD.
Wilde & Welch (2022)	Attention deficit hyperactivity disorder (ADHD) and musical behavior: The significance of context	The research results indicate that, despite ADHD-related symptoms, affected individuals could successfully participate in music and acquire musical skills. ADHD was not a static condition but varied according to the pedagogical and musical context.
Grob <i>et al.</i> (2022)	Musical Performance in Adolescents with ADHD, ADD, and Dyslexia—Behavioral and Neurophysiological Aspects	Compared to dyslexic adolescents, controls, as well as adolescents with ADHD and ADD, performed better in rhythmic reproduction, rhythmic improvisation, and musical expression after intervention. A better understanding of these differences may help to develop personalized therapeutic or prevention interventions.
Anand (2022)	The Potential of Music Training to Improve Attentional Control and Inhibitory Control in Children with ADHD	The literature reviewed suggests that music training has the potential to be an effective, safe, stigma-free, and long-term intervention for improving attentional and inhibitory control in children with ADHD.
Ramaswamy <i>et al.</i> (2024)	Therapeutic use of music in neurological disorders: A concise narrative review	The impact of music in influencing neuroplasticity in multiple brain regions/systems, the development, and optimization of mental, emotional, behavioral, sleep, and cognitive functions, as well as increasing quality of life, reveals its opportunity to be harnessed as a potent non-invasive, non-pharmacological treatment alternative.
Branctisano <i>et al.</i> (2020)	Why is music therapeutic for neurological disorders? The Therapeutic Music Capacities Model	Existing research findings are consistent with the therapeutic music capacity model (TMCm), and demonstrate that each of the seven music capacities, employed in various combinations and contexts, can achieve behavioral, cognitive, psychosocial, and motor benefits.
Kamioka <i>et al.</i> (2014)	Effectiveness of music therapy: a summary of systematic reviews based on randomized controlled trials of music interventions	Treatment with MT improved global and social functioning in schizophrenia or severe mental disorders, gait and activities related to Parkinson's disease, depressive symptoms, and sleep quality. No specific adverse effects or harmful phenomena occurred in any of the studies, and almost all patients tolerated MT well.

Note. Authors' development

Discussion

Effects on hyperactivity

First, Situmorang et al. (2018) approach music therapy or music therapy in two approaches: passive or receptive, and active or creative. In that sense, they highlight the benefits of the active nature which leads to an externally visible action that is reflected in the patient's behavior on his own initiative, such as improvisation, which leads him to improve the control of hyperactivity. Due to the sample size employed in this study of 14 millennial students of BK Unika Atma Jaya with a high level of academic anxiety, it is hoped that future researchers can evaluate the effectiveness of group cognitive behavioral therapy with passive and active music therapy techniques using a real experimental design, with random assignment in each group to obtain two equal groups and representative sample size.

Acebes-De Pablo & Carabias-Galindo (2016) contribute that music education provides a suitable interdisciplinary space to organize the development of musical competencies with treatment oriented to children with ADHD educational support through music therapy, also contributing to the control of hyperactivity.

In the case of Luo et al. (2023), in addition to music therapy, they added yoga as a spatial activity, since it provides very positive effects in terms of improvement in attention and control of hyperactivity and impulsivity in children. The results have a high practical value, where schools can organize yoga and music programs according to the characteristics of their children, and use classroom or extracurricular activities to provide these complementary forms of therapy. Even so, research limitations are evident such as the use of a single tool such as the ADHD Rating Scale MTA SNAP-IV, which suggests the use of various instruments for future studies. Similarly, the research question addressed was not clearly answered, because the current results do not address this issue and very few researchers have delved into it. It is left as a gap for future research.

In the research context of Wilde & Welch (2022), the usual symptoms of ADHD are reduced or disappear completely when participants play and perform music. Analyses suggest that effective, context-sensitive, inclusive pedagogy can integrate students into successful individual and collective music-making, and thus progressively control hyperactivity and impulsivity.

Specifically, rhythm, melody and tempo are tools used to address non-musical behaviors and to drive whole-body changes. Much of this research shows that pleasurable music increases dopamine levels in the brain. This neurotransmitter, responsible for regulating attention, working memory and motivation is in short supply in people with ADHD, so music therapy represents a positive alternative. This has been reinforced by studies conducted by Blasco-Magraner et al. (2021) and Reybrouck et al. (2021), who propose a new approach to music listening that integrates arguments from different disciplines and converges in the construction of an integral aesthetic perspective, capable of elegantly maneuvering between biological and cultural dimensions. However, this approach also opens new perspectives for future research.

Music therapy, comprehension, and concentration

For Jacob & Pillay (2021), music therapy improved the reading skills of students with intellectual disabilities, therefore, it can be adapted for complementary treatment in children and youth with ADHD to improve their concentration. Nevertheless, it is necessary that in future similar studies, the sample size be established by means of a statistical method, and in this way, results with a greater research projection will be obtained. However, Mendes et al. (2024) did not have the same favorable result, because in their experiments they did not manage to associate music therapy with significant changes in the children and youngsters who were part of their study, which disagrees with what was contributed by Cibrian et al. (2022) and León et al. (2024), on the importance of achieving socio-communicative and emotional regulation in children and adolescents, as well as attention, memory and executive function.

It should be noted that in contrast, Dursun et al. (2021), statistically, failed to test the established hypotheses associating music with the improvement of ADHD symptoms. Although they used instruments such as the CPT (MOXO d-CPT), the Conners Rating Scales for Parents (CPRS) and Teachers (CTRS), and the ADHD Self-Report Scale for Adults (ASRS) according to the ages of the participants, in addition to the listening checklist, they failed to test under the conditions of this study the established premises; however, they recommended further studies by modifying the conditions in future experiments. According to the authors, low frequency and slow rhythm music could be more useful for auditory training, while in cases of greater difficulties in receptive and expressive listening, medium and high frequency musical content should be prioritized. They also suggest reviewing the characteristics and conditions of the three cases addressed and adapting the environments for future studies.

For their part, Madjar et al. (2020), in their experimental work, evaluated in-depth reading comprehension and mean levels of changes in heart rate variability (before and during each task) using validated instruments. When comparing over time, they found that reading comprehension improved significantly as a result of the musical conditions in the ADHD group. The differences in the changes in heart rate variability were significant between the groups, which allowed them to explain the reading performance of the members.

It is important to highlight that Martínez-Vérez et al. (2024) emphasized the improvement of concentration in school-age children, which reflects the relevance of these therapies during a vital period for interpersonal and cognitive development. This justifies the integration of music therapy and art therapy in educational and therapeutic programs for children with these conditions.

Regarding the experimental contributions of Grob et al. (2022), adolescents with ADHD evidenced better performance in rhythmic reproduction, rhythmic improvisation, and musical expression, compared to groups of students with dyslexia. This was performed by music experts considering aspects such as rhythmic and pitch accuracy, intonation, improvisation skills, and musical expression, which represented an improvement in the concentration of these students.

On the other hand, evidence collected by Anand (2022) shows how music training has improved attention, concentration, and inhibition in neurotypical individuals. Likewise, in those with neurodiverse disorders, it provides promising opportunities for future research that may explore how music training can improve executive functions in children with ADHD. As this is a literature review, considering 15 studies looks like an unrepresentative sample to be conclusive. This highlights the need for further research comparing musical interventions, non-musical skill development interventions, and non-musical educational interventions to assess whether music training specifically impacts these outcome variables or whether any intervention can generate such improvement. Also, whether music training improves specific types of attention rather than attention in general should be investigated.

Brancatisano et al. (2020) make the thoughtful point that a variety of psychological and neural mechanisms underlie the benefits of music and music-based therapies. These range from auditory-motor coupling, neuroplasticity, activation of the mirror neuron network (MNS), facilitation of neurobiological and hormonal systems, as well as increased arousal by elevating the person's mood, attention, and concentration.

In this section, it is appreciated that music therapy can improve mental and emotional states, alter the mood of children and youth, regulate thought patterns, and enrich the cognitive modes of the brain. Personalized music education proves to be effective and practical in reducing stress during learning and in crisis intervention. This agrees with the contributions of Langberg et al. (2023) and Rodwin et al. (2023) who state that, by combining the characteristics and effects of music with the theory of music-assisted therapy for children with ADHD, it generates important contributions on comprehension and concentration.

Effects on self-control and social interaction

In this line of ideas, Martin-Moratinos et al. (2023) agree that from the contributions detected, active music therapy improves hemispheric synchrony, social skills, aggressiveness, and impulsivity. Passive music therapy improves academic skills such as arithmetic, drawing, and reading comprehension, as well as attention and disruptive behaviors.

For their part, Park et al. (2023) propose a new alternative to medicine for the prevention and treatment of depression through various uses of music therapy in children and adolescents with ADHD. The study by Lee et al. (2023) indicates that intervention with the use of music and movement is a potential alternative and effective tool for the treatment of ADHD and can significantly improve the quality of life and attention of patients. However, the scope of this study is limited because of the non-significant sample size. Likewise, only children with ADHD-I or ADHD-C were considered, so the conclusions are not applicable or generalizable to all ADHD subtypes. Finally, the music used in this study was children's songs. The effects of other types of music on people with ADHD warrant further investigation. While Llamas (2014) and Khaleghian et al. (2023), through both experimental and quasi-experimental studies, highlight that this type of alternative therapy improves the interaction capacity, social skills and psychological health of hyperactive children and youth with their peers.

Similarly, Hemmati et al. (2016) highlighted because of their experimental study, that the effectiveness of group art therapy includes musical activities to improve the self-control of students with ADHD. This therapy can be used by teachers, parents, or professionals working with such students. In the same context, Kim & Wheeler (2023) noted that music therapy, with the management of melodic contours, has shown promise as a form of treatment for individuals affected by neuropsychological behaviors for attention needs and demonstrations of impulsive dyscontrol, which are associated with ADHD.

Shuman et al. (2016) documented through a preliminary retrospective study the immediate impact of 60- to 90-minute group music therapy sessions on mood in adolescent patients diagnosed with a variety of psychiatric disorders associated with ADHD. They were supported by the application of a binary outcome model of change or absence of change in the mood of individuals. Because of the positivity of their findings, future research on the benefits of music therapy for adolescents suffering from psychiatric disorders is expected to be supported.

On the other hand, Kamioka et al. (2014) and Martínez-Vérez et al. (2024), after an exhaustive review of the updated literature on the subject, specified that the results describe the efficacy and favorable perception of music therapy, as well as

art therapy in the treatment of children and young people with schizophrenia or severe mental disorders, gait and related activities in Parkinson's disease, depressive symptoms and sleep quality, as well as attention deficit hyperactivity disorder, mainly for the treatment of deficiencies associated with cognitive, emotional and social skills.

For Chenguang (2021), experimental results have corroborated that music therapy combined with cognitive-behavioral intervention can improve the cognitive functions of children with ADHD, in addition to having clinical application values. This is supported by the comparative results of the before and after intervention performed on diagnosed children, who significantly improved their concentration by reducing the percentage of errors in the assignments performed.

In a special direction, Ramaswamy et al. (2024), through a comprehensive review of the available literature regarding the benefits of music therapy, determined that this form of therapy is effective in a variety of conditions, including those that damage the developing brain, such as autism and ADHD. Significantly, this research shows that the relationship between music and the ADHD brain is likely explained by the theory of selective attention load. According to this theory, focusing on a task involves balancing two cognitive mechanisms, such as early selection, i.e., a deliberate choice of a target stimulus with a specific goal, and late selection, i.e., involuntary attention attracted to a dominant stimulus. Taken together, music-based interventions result in appreciable beneficial effects on self-regulation in ADHD subjects.

It should be noted that in the study by Ramaswamy et al. (2024) no specific adverse effects were observed in any of the cases and almost all patients tolerated music therapy significantly. Despite the possible multimodal therapeutic aspects against different neurological disorders, there is insufficient evidence on the exact nature of music therapy. As a relevant fact, a limitation in the studies analyzed was the absence of standardized protocols, the consideration of sociocultural aspects and the environmental conditions of the place. The central neurophysiological mechanism underlying the therapy and the effect of music should be explored in as much detail as possible to better understand the pathways involved, although this may be difficult in clinical and preclinical settings.

In most of the studies reviewed, no definitive effects of music therapy for people with ADHD were detected. However, some results suggest that a music therapy program that promotes autonomy and creativity may help children and adolescents to interact more appropriately with others in a residential setting. Likewise, it could lead to a slight temporary increase in disruptive behavior in the classroom. These aspects align with the contributions of Leon et al. (2024) and Cibrian et al. (2022), who point out that a more structured program working with smaller groups may be advantageous for children with ADHD. Similarly, music can help children with special needs integrate into group settings when appropriate intervention measures are implemented.

FINAL REMARKS

The findings of the analyzed research show that the benefits of music in the treatment of ADHD range from the improvement of social relationships and communication, the contribution to relaxation and reduction of anxiety levels, and the improvement of attention and concentration in children and young people with this disorder.

Despite having at hand, a considerable amount of favorable evidence on the use of music as part of non-pharmacological therapies in patients diagnosed with ADHD, it still cannot be seen as an exclusive clinical treatment. So far, it has only been used as a complementary method that has been developed through some techniques, such as the use of auditory means of musical expansion, musical training in the handling of instruments, the use of synchronized body movements, instrumental improvisation, singing and relaxation activities both in groups and individually.

The present study identified limitations that should be considered in future research. Firstly, it is suggested that the number of documents analyzed should be increased to broaden the scope of the findings and complement the critical analysis. It is also recommended that priority be given to the inclusion of empirical, quantitative or mixed-approach studies that include both experimental and control groups, with significant sample sizes determined by means of rigorous statistical procedures. In addition, it is advisable to use two or more standardized techniques and instruments to enrich the research contribution.

We emphasize the importance of these studies being supported by a literature review that substantiates the need for the research, particularly in nations located in low-income regions, to facilitate obtaining sponsorship from international organizations dedicated to the study of ADHD.

Despite the multimodal therapeutic potential of music therapy for various neurological disorders, including ADHD, most of the studies reviewed do not provide sufficient or conclusive evidence on the exact nature of the musical intervention or its effects on comprehension, concentration, self-control and social interaction. However, some experimental studies stand out, such as that of Park et al. (2023), focused on the prevention and treatment of depression through different modalities of music therapy in children and adolescents with ADHD. Since this study was conducted from a neurophysiological perspective, it is recommended that future research incorporate other dimensions and techniques that have shown more conclusive

results.

Regarding the studies analyzed, it was observed that, in several of quantitative approach with experimental or observational design, the sample size was not calculated following statistical protocols, so it was not significant. Likewise, in three of these papers, the assignment of participants to the experimental and control groups was not randomized, increasing the risk of bias. Another aspect to highlight is that, in some music therapy interventions, a standardized procedure was not applied for the selection of the type of music, nor were the environmental conditions adequately controlled.

Although this study identified works with well-detailed and well-founded materials and methods, future researchers are urged to precisely delimit the object of study, based on a thorough review of updated and available literature. Likewise, the importance of framing research on music therapy and ADHD as an opportunity to provide robust results, applying standardized methods, regardless of the methodological approach adopted, is emphasized.

From a societal point of view, it is advocated that the international scientific community should promote research in low-income countries, with the aim of fostering collaboration between governments, non-governmental organizations and pharmaceutical companies, and thus ensure that ADHD treatment is affordable and widely available. Similarly, it is proposed that future research should integrate mental health services into primary health care systems, to analyze and implement practical interventions that further explore the therapeutic benefits of music in children and adolescents with ADHD, as well as the use of technological resources for its development.

Finally, it is stated that “the application of music therapy designed by specialists for children and adolescents with ADHD generates positive results, improving the level of concentration, self-esteem, the progressive development of social skills, the control of hyperactivity and the reduction of anxiety states”. This statement is proposed as a specific starting point for the continuity of new and innovative scientific studies that contribute to the well-being of people with this condition.

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Task	% of contribution of each author				
	A1	A2	A3	A4	A5
A. theoretical and conceptual foundations and problematization:	20%	20%	20%	20%	20%
B. data research and statistical analysis:	20%	20%	20%	20%	20%
C. elaboration of figures and tables:	20%	20%	20%	20%	20%
D. drafting, reviewing and writing of the text:	20%	20%	20%	20%	20%
E. selection of bibliographical references	20%	20%	20%	20%	20%
F. discussion of results and concepts	20%	20%	20%	20%	20%

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